

Credit Card Fraud Detection: A Comparative Study of Anomaly Detection and Supervised Learning under Extreme Class Imbalance

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Abstract—The severe class disparity and changing assault patterns make it difficult to identify credit card fraud. This paper compares a supervised baseline (XGBoost) with three unsupervised anomaly detection techniques (Isolation Forest, One-Class SVM, and an Autoencoder) on a severely skewed real-world dataset (0.17% fraud). While XGBoost uses labeled data with class weighting, models are trained under a realistic protocol where anomaly detection techniques only learn from typical transactions. Precision, recall, F1-score, and PR-AUC are used to assess performance. The results suggest that XGBoost outperformed unsupervised approaches, which had good recall but low precision, by achieving the highest F1-score (0.64). The results show that supervised and unsupervised methods in fraud detection systems have trade-offs.

Index Terms—Fraud Detection, Machine Learning, Anomaly Detection, XGBoost, Autoencoder

I. INTRODUCTION

Because credit card fraud is uncommon and constantly changing, detecting it is a significant difficulty. Conventional evaluation techniques frequently depend on precision, which makes them unsuitable for datasets with significant imbalances. To properly evaluate model performance, this study focuses on precision-recall metrics, especially PR-AUC.

The main contributions of this study are:

- A fair comparison of supervised and unsupervised models under identical conditions
- Evaluation using precision-recall metrics for imbalanced data
- Analysis of threshold sensitivity in anomaly detection techniques

II. DATASET

Of the 284,807 transactions in the sample, 492 (0.17%) are fake. PCA transformation is used to anonymize features, along with other characteristics like Amount and Time. Time is not included in the study, but the Amount feature is standardized. The adoption of evaluation criteria based on precision-recall is motivated by the significant imbalance.

III. METHODOLOGY

A. Isolation Forest

Isolation Forest isolates anomalies based on random partitioning. Anomalies tend to have shorter path lengths.

B. One-Class SVM

One-Class SVM learns a boundary around normal data and is sensitive to hyperparameters and feature scaling.

C. Autoencoder

The Autoencoder reconstructs input data. High reconstruction error indicates anomalous transactions.

D. XGBoost

XGBoost is used as a supervised baseline with class imbalance handled using `scale_pos_weight`.

IV. EXPERIMENTAL SETUP

An 80/20 stratified method is used to divide the dataset into training and testing sets. The test set includes the entire distribution, however anomaly detection models are only trained on typical transactions.

For One-Class SVM and Autoencoder models, StandardScaler is used to prevent data leaking. Precision, recall, F1-score, and PR-AUC are evaluation metrics.

V. RESULTS

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Precision	Recall	F1-score
Isolation Forest	0.097	0.643	0.168
One-Class SVM	0.121	0.837	0.211
Autoencoder	0.047	0.827	0.090
XGBoost	0.512	0.867	0.644

VI. DISCUSSION

The findings show that XGBoost's ability to use labeled data and successfully manage class imbalance allowed it to attain the greatest overall performance.

By identifying non-linear patterns in typical transactions, the autoencoder was able to attain high recall, but it suffered from low precision because of an increase in false positives.

The high-dimensional feature space and sensitivity to hyperparameters caused One-Class SVM to struggle. Although it was sensitive to contamination settings, Isolation Forest performed mediocrely.

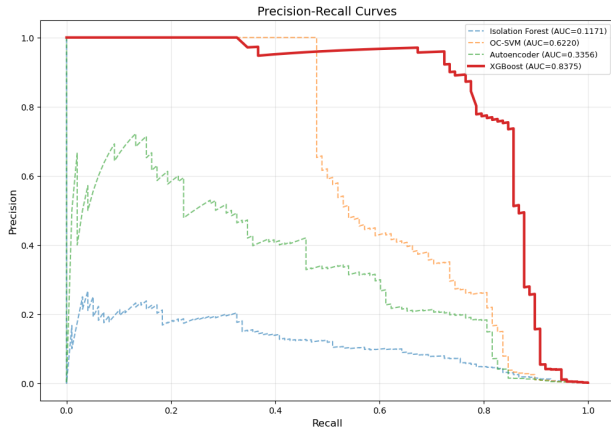


Fig. 1. Precision-Recall Curve

VII. CONCLUSION

This study shows that when labeled data is available, supervised models perform better than unsupervised approaches. However, in situations where labeled data is scarce, anomaly detection methods are still helpful. Future research might investigate hybrid models that combine the two methods.

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